

CareBridge: Intelligent Elder Support & Care Prioritization System

Problem Statement

Countries like Japan are facing a severe caregiver shortage due to an ageing population. At the same time, not every elderly person requires the same level of support.

Current solutions either:

- focus only on emergencies,
- monitor physical health,
- or provide the same type of support to everyone.

However, the real challenge is identifying who truly needs caregiving support while enabling others to remain independent.

Our Solution

CareBridge is a preventive elder support ecosystem that:

- Tracks changes in independence.
- Prioritizes caregiving resources for those who need them the most.
- Encourages social well-being.
- Empowers elderly individuals to seek help easily.

App Features

1. Independence Dashboard

Tracks trends in:

- Mobility
- Hand movement/activity
- Social engagement

2. Care Priority Score (MOST IMPORTANT)

Score	Interpretation
80–100	Independent
60–79	Periodic monitoring
40–59	Additional support may be beneficial
Below 40	High-priority caregiving recommended

3. One-Tap Assistance

This is for elderly users who need help but hesitate to ask.

Interface:

Need Help Today?

- Groceries
- Transportation
- Medicine Pickup
- Talk to Family
- Request Caregiver Visit

When pressed:

- Family members get notified.

OR

- Local caregiver/community volunteer gets notified.

4. Suggestion Section

Based on trends, the app provides recommendations.

If mobility decreases:

"Your walking activity has reduced recently. Consider taking short daily walks or discussing this change with your healthcare provider."

If social engagement decreases:

"We've noticed fewer social interactions this week. Connecting with family or friends can positively support well-being."

Buttons:

- Call Daughter
- Call Son
- Join Community Activity

If independence remains stable:

"Great job! You've maintained your activity levels this week."

5. Family Dashboard

Family members can view:

- Weekly reports
- Assistance requests
- Care Priority Score
- Wellness insights

6. Weekly Wellness Reports

Example:

Mobility:

Walking consistency decreased by 12%.

Social Engagement:
Speaking duration decreased by 18%.

Assistance Requests:
Groceries requested twice.

Recommendation:
Additional support may be beneficial.

Hardware: Smart Neckband

Why neckband?
Because it allows speech activity estimation while remaining comfortable.

Components

1. ESP32
Acts as the controller.

Functions:
• Processes sensor data.
• Sends data to the app via Bluetooth.

2. Throat Microphone (Laryngophone)

Detects:
• Vocal cord vibrations.

Used for:
• Estimating speaking duration.

NOT used for:
• Recording conversations.
• Storing speech content.

3. MPU6050

Detects:
• Walking patterns.
• Activity consistency.
• Hand movement trends.

High-Priority Care Alert

If the Care Priority Score continues to decrease over multiple weeks, indicating a consistent decline in mobility, social engagement, and independence, then every time the elderly person opens the app, a full-screen notification will appear, stating:

"You now need a caregiver."

Below this message, the following options will be displayed:

- Request Caregiver
- Notify Family Members
- Share Wellness Report